NEURAL NETWORKS

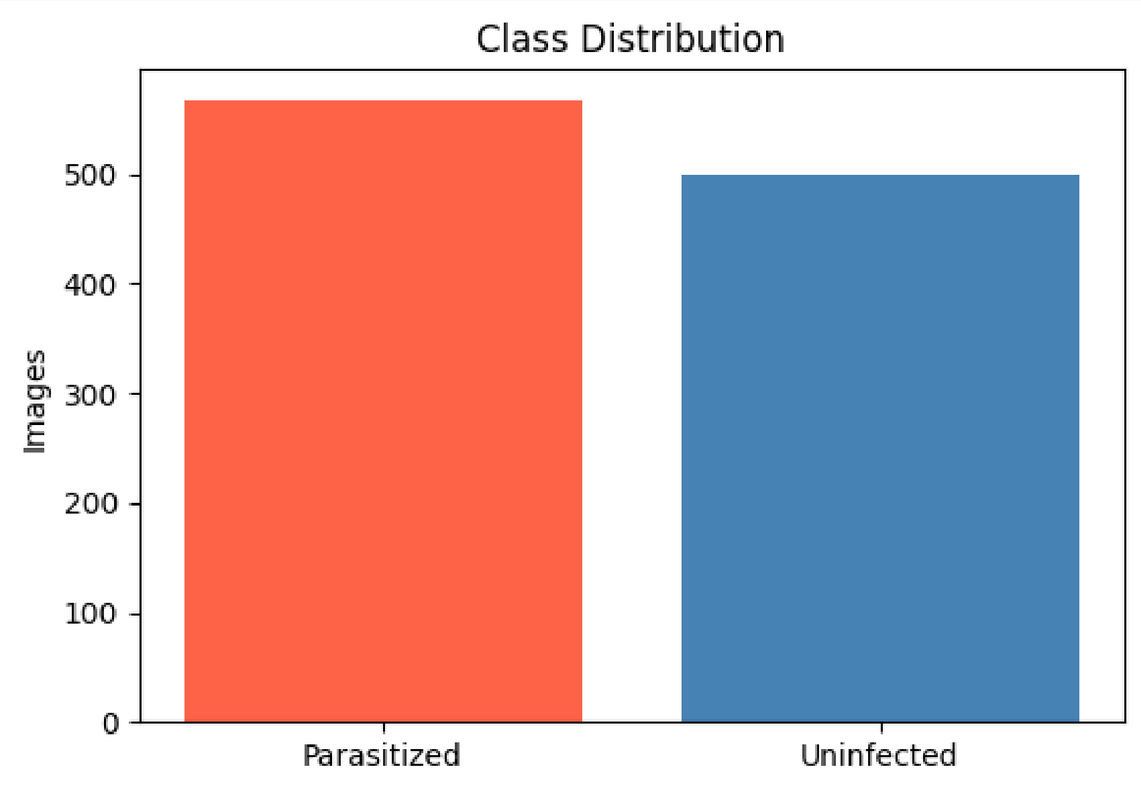
DANIEL TROYA

Malaria

INFECTED CELL IMAGE CLASSIFIER

1 | DATASET MANIPULATION

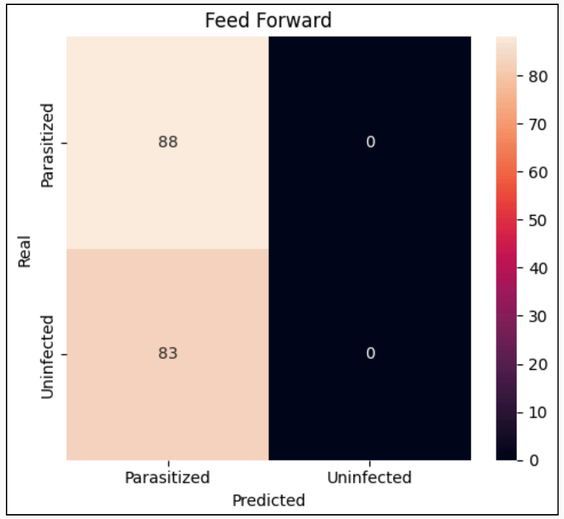
Model	Normalization (0-1)	VGG16 Normalization	One-hot Encoding
FNN 64 x 64	O	X	O
CNN 64 x 64	O	X	O
CNN + Aug 64 x 64	O	X	O
VGG16 + RF 224 x 224	X	O	O
VGG16 + DL 224 x 224	X	O	O

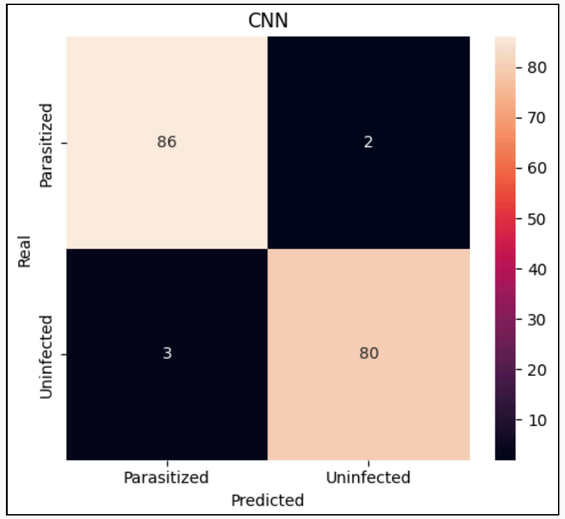


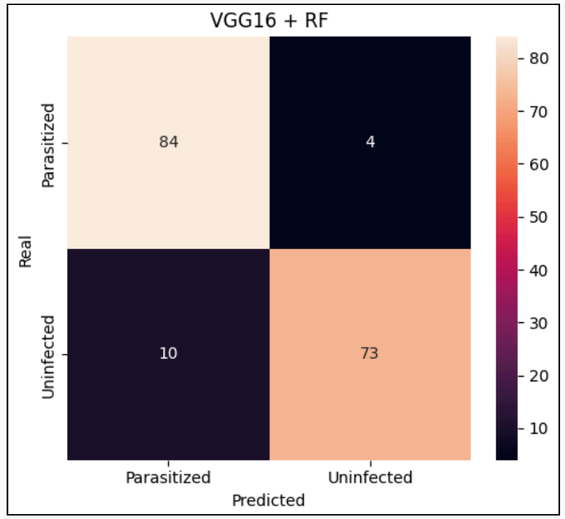
- One-hot encoding for all models
- Use stratification to maintain the same class distribution:
  - train(80%) → train(80%) and validation(20%)
  - test (20%)

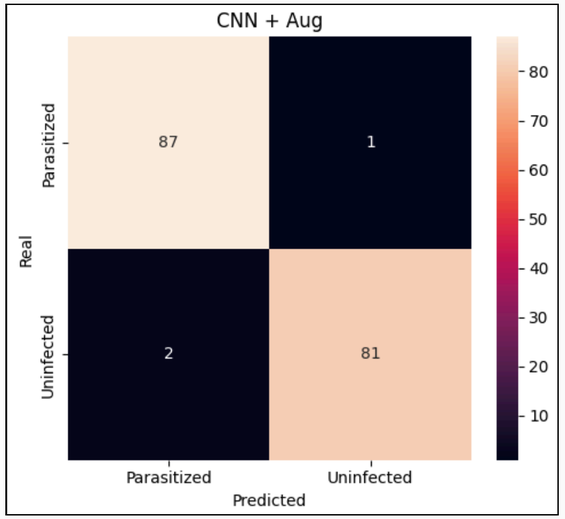
- Data label are balancing
- 500 Parasitized - label = 1
- 567 Uninfected - label = 0

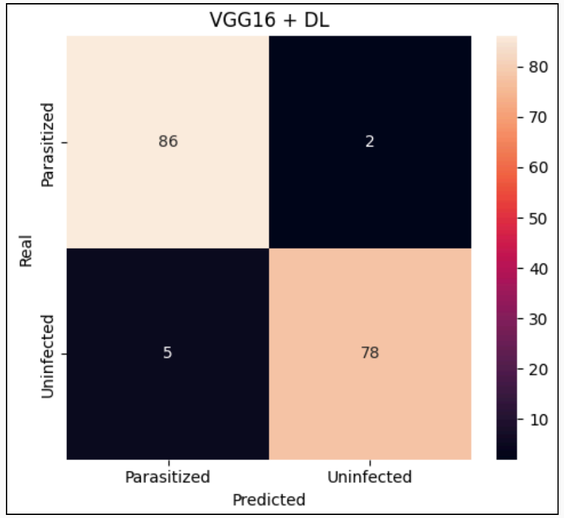
Features / Models	Feed Forward	CNN	CNN + Aug	VGG16 + RF	VGG16 + DL
Architecture	***Input layer:*** 12288 n ***Hidden layer 1:*** 256 n (ReLU) ***Hidden layer 2:*** 128 n (ReLU) ***Hidden layer 3:*** 64 n (ReLU) ***Output layer:*** 2 n (Softmax)	***Input layer:*** 12288 n ***Conv:*** 32 fil → MaxPool → Dropout ***Conv:*** 64 fil → MaxPool → Dropout ***Conv:*** 128 fil → MaxPool → Dropout ***Flatten:*** 4608 n ***Dense:*** 128 n ***Output layer:*** 2 n (Softmax)		***Input layer:*** 150528 n ***Conv:*** 64 fil → MaxPool ***Conv:*** 128 fil → MaxPool ***Conv:*** 256 fil → MaxPool ***Conv:*** 512 fil → MaxPool ***Conv:*** 512 fil → MaxPool ***Flatten:*** 25088 n ***Feature Extractor*** ***Output layer:*** RF, DL (2 classes)	
Optimizer	Adam	Adam	Adam	-	Adam
Loss	Categorical Crossentropy	Categorical Crossentropy	Categorical Crossentropy	Gini Impurity	Categorical Crossentropy
Early Stopping	No	Yes (patience 5)	Yes (patience 10)	-	Yes (patience 10)
Epochs - Batch size	50 - 16	60 - 16	60 - 32	-	60 - 16
Feature Extractor	-	-	-	Random Forest	Deel Learning (256,128,64 ReLU)
Model Size	36.51 MB	7.87 MB	7.87 MB	0.38 MB	74.01 MB











MODEL	ACCURACY	PRECISION	AUC
CNN + Aug	0.9825	0.9878	0.9993
CNN	0.9708	0.9756	0.9979
VGG16 + DL	0.9591	0.9750	0.9975
VGG16 + RF	0.9181	0.9481	0.9862
FNN	0.5146	0	0.5

Metrics calculated on validation set

Top 3 Ranked by Validation AUC

1. CNN + Aug → 0.9993
2. CNN → 0.9979
3. VGG16 + DL → 0.9975

Why AUC? Threshold-independent — more reliable than accuracy for medical classification.

The Top 3 models are combined using a weighted average to obtain a stronger final model:

- Weights range from 0.0 to 1.0 (steps of 0.1)
- Weights must sum to 1.0 → to preserve valid probability distributions
- Best combination selected by highest AUC on validation:
  - CNN + Aug: 0.5
  - CNN: 0.0
  - VGG16 + DL: 0.5

Final Ensemble Results

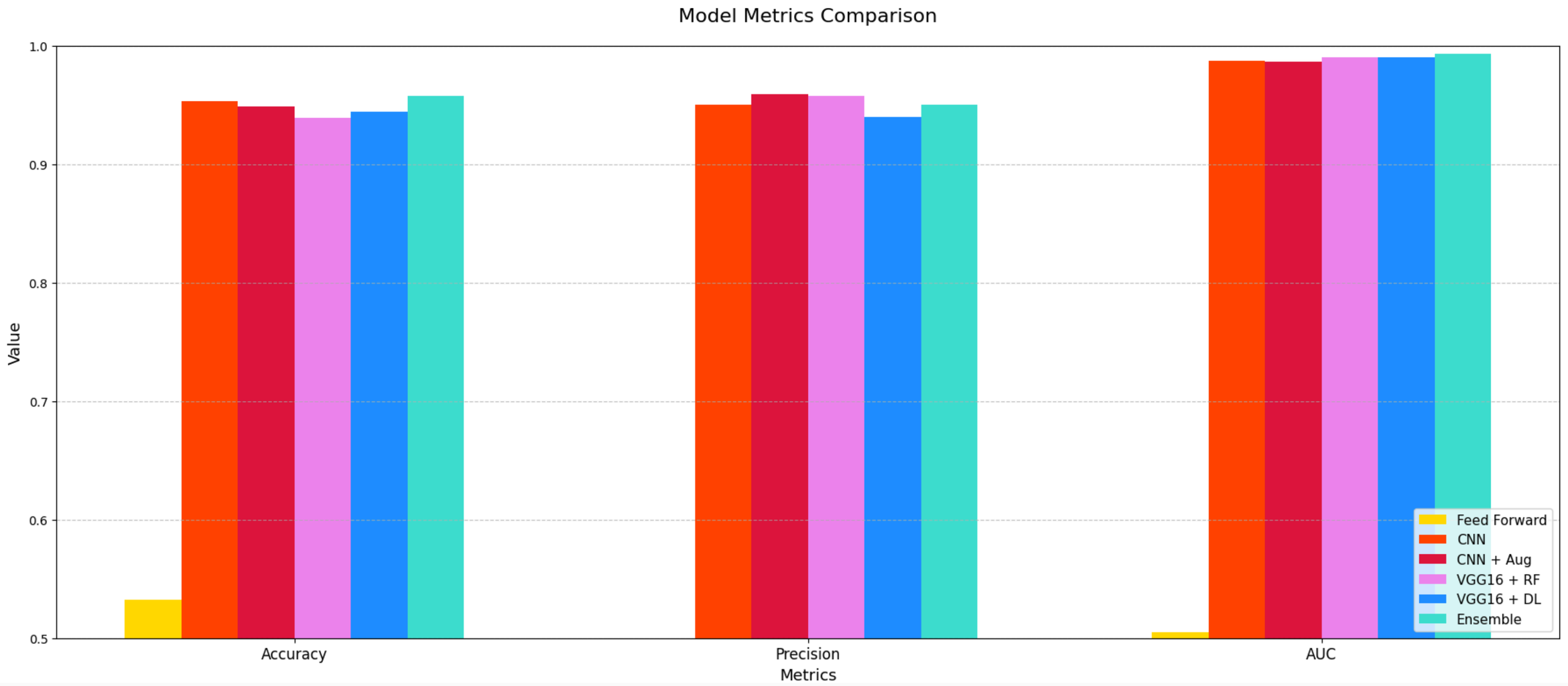
$\hat{y}$ = 0.5 × P(CNN+Aug) + 0.0 × P(CNN) + 0.5 × P(VGG16+DL)

$\hat{y}$ = 0.99 (AUC Ensemble in Validation)

MODEL	CNN + Aug	CNN	VGG16 + DL
SIZE	7.87 MB	7.87 MB	74.01 MB

Total Esemble Size: 89.74 MB

5 | FINAL RESULTS



The best model was the Ensemble, which combined the Top 3 models (CNN+Aug, CNN, and VGG16+DL) to achieve superior performance with an AUC of 0.9932 on the test set. VGG16 models were competitive but showed only marginal differences compared to CNN-based models. The Feed Forward Network was the worst model — the only one unable to capture spatial relationships in cell images, resulting in an AUC of approximately 0.50. Although the Ensemble achieved the highest AUC, its total size of 89.74 MB makes it unsuitable for mobile deployment, so CNN+Augmentation was selected instead, offering strong performance (AUC 0.9868) at only 2.61 MB after TFLite conversion.

DANIEL TROYA

github.com/dtpollo

Thank You